for Forward kinematics

Parameter table

θ	α	r	d
θ_5	$90°$	a	0
θ_4	0	b	0
θ_3	0	c	0
θ_2	$-90°$	d	0
θ_1	0	0	e

$$H_1^0 = \begin{bmatrix} \cos\theta_5 & 0 & \sin\theta_5 & a\cos\theta_5 \\ \sin\theta_5 & 0 & -\cos\theta_5 & a\sin\theta_5 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$H_2^1 = \begin{bmatrix} \cos\theta_4 & -\sin\theta_4 & 0 & b\cos\theta_4 \\ \sin\theta_4 & \cos\theta_4 & 0 & b\sin\theta_4 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$H_3^2 = \begin{bmatrix} \cos\theta_3 & -\sin\theta_3 & 0 & c\cos\theta_3 \\ \sin\theta_3 & -\cos\theta_3 & 0 & c\sin\theta_3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$H_4^3 = \begin{bmatrix} \cos\theta_2 & 0 & -\sin\theta_2 & d\cos\theta_2 \\ \sin\theta_2 & 0 & \cos\theta_2 & d\sin\theta_2 \\ 0 & -1 & 0 & d \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$H^4_5 = \begin{bmatrix} \cos\theta_1 & -\sin\theta_1 & 0 & 0 \\ \sin\theta_1 & \cos\theta_1 & 0 & 0 \\ 0 & 0 & 1 & e \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\underline{H^0_5 = H^0_1 \times H^1_2 \times H^2_3 \times H^3_4 \times H^4_5}$$